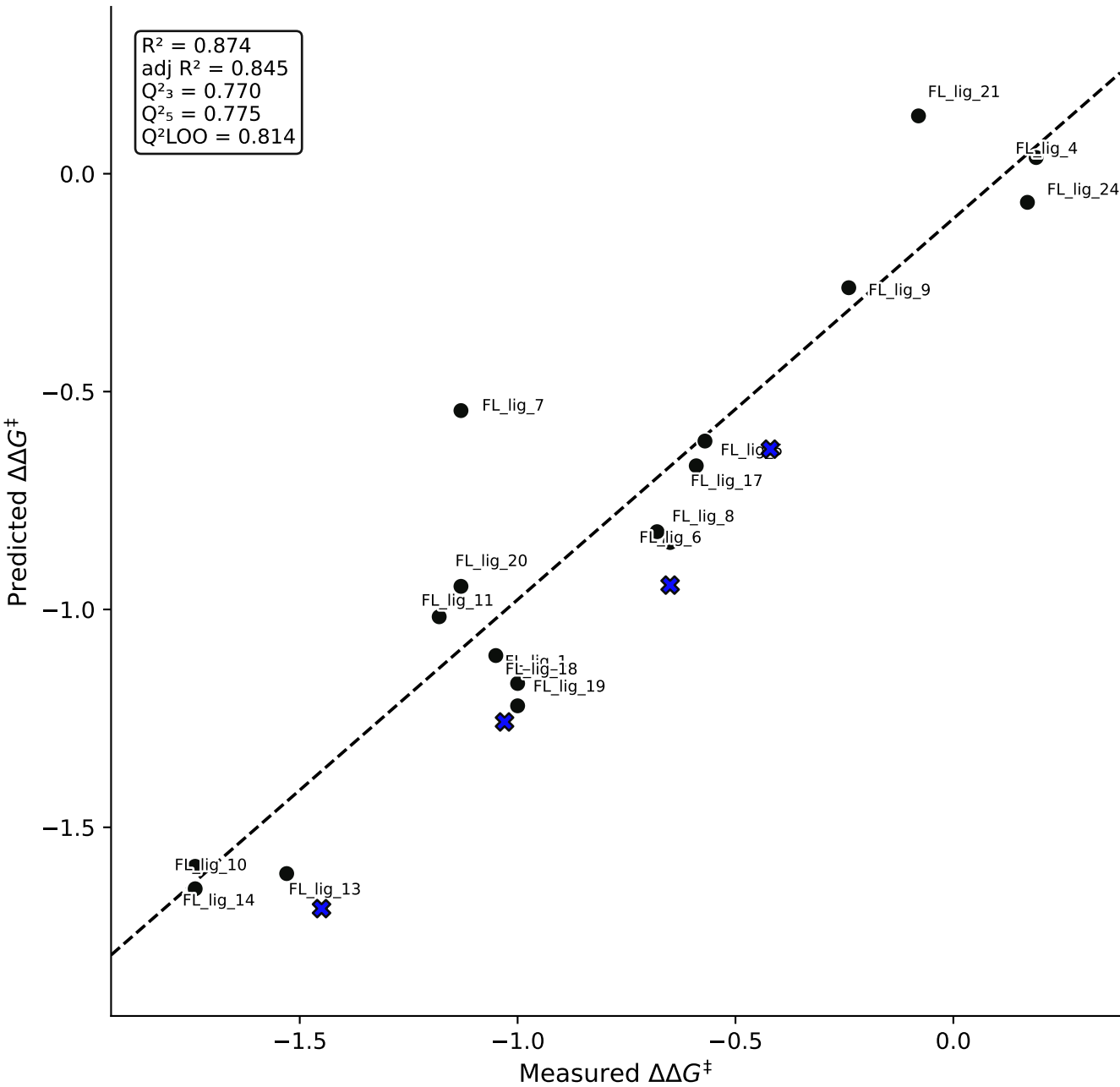


Predicted vs Measured



Model Metrics

| Metric                | Value  |
|-----------------------|--------|
| $R^2$ (train)         | 0.8743 |
| adj $R^2$ (train)     | 0.8453 |
| MAE (train)           | 0.1641 |
| $Q^2$ 3-fold          | 0.7698 |
| MAE 3-fold            | 0.2291 |
| $Q^2$ 5-fold          | 0.7749 |
| MAE 5-fold            | 0.2287 |
| $Q^2$ LOOCV           | 0.8140 |
| MAE LOOCV             | 0.2080 |
| $R^2$ (ext. val.)     | 0.7605 |
| adj $R^2$ (ext. val.) | 0.0419 |
| MAE (ext. val.)       | 0.2885 |

Coefficient Estimates

| Estimate | Std. Error | t value | p value |
|----------|------------|---------|---------|
| -0.917   | 0.058      | -15.822 | 0.0     |
| 0.266    | 0.06       | 4.403   | 0.001   |
| 0.386    | 0.056      | 6.924   | 0.0     |
| 0.136    | 0.061      | 2.236   | 0.043   |

Cross-Validation Metrics

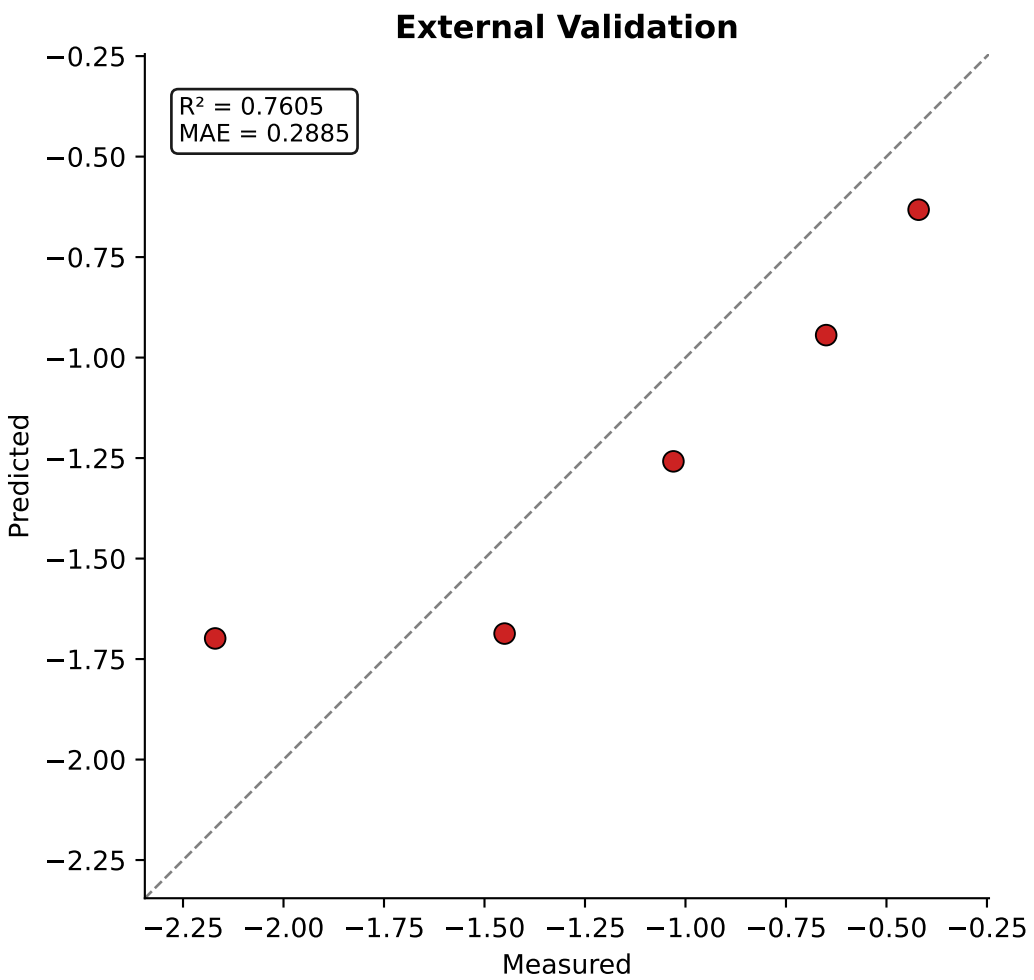
| Metric | 3-Fold | 5-Fold | LOOCV  |
|--------|--------|--------|--------|
| Q²     | 0.7698 | 0.7749 | 0.814  |
| MAE    | 0.2291 | 0.2287 | 0.208  |
| RMSD   | 0.2768 | 0.2727 | 0.2494 |

Variance Inflation Factors (VIF)

| variable   | VIF   |
|------------|-------|
| B1_6-4     | 1.07  |
| loc_B5_6-7 | 1.064 |
| loc_B5_6-4 | 1.006 |

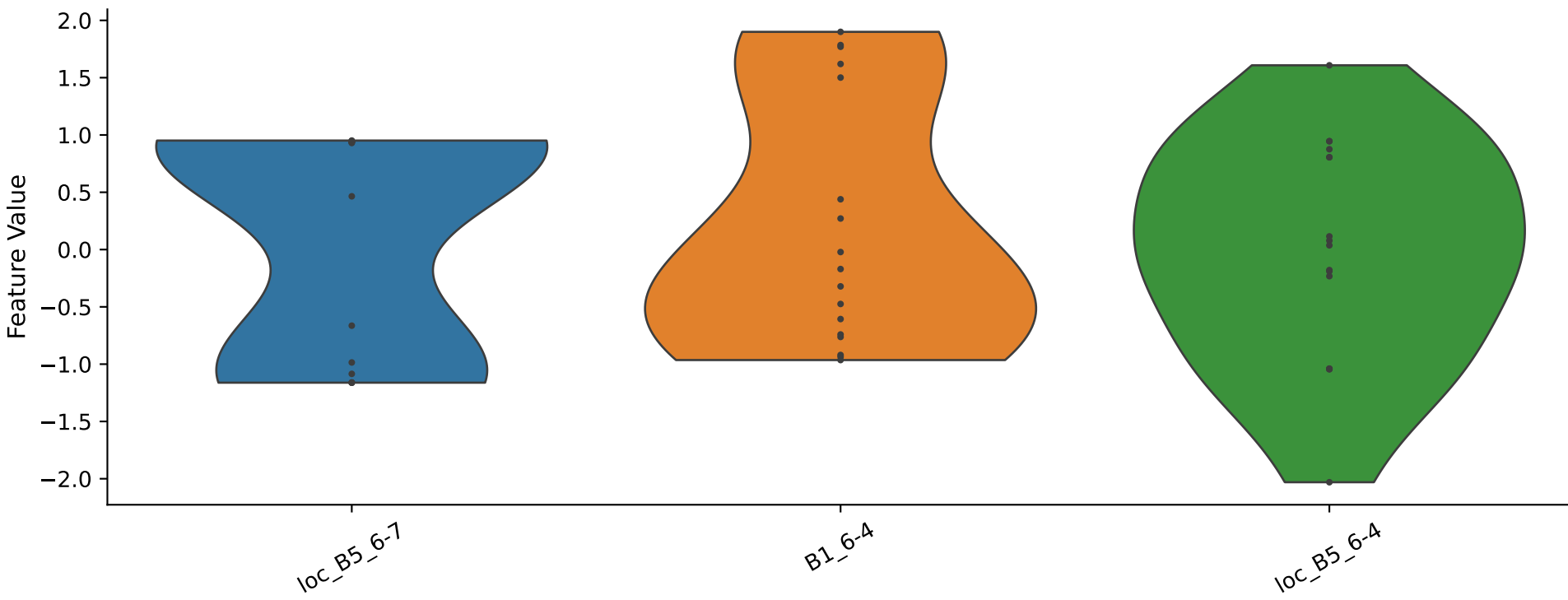
Coefficient Estimates

| Estimate | Std. Error | t value | p value |
|----------|------------|---------|---------|
| -0.9168  | 0.0579     | -15.822 | 0.0     |
| 0.266    | 0.0604     | 4.4033  | 0.0007  |
| 0.3861   | 0.0558     | 6.9236  | 0.0     |
| 0.1357   | 0.0607     | 2.2362  | 0.0435  |



| Molecule  | Actual | Predicted | Error in % |
|-----------|--------|-----------|------------|
| FL_lig_3  | -0.65  | -0.944    | 45.25      |
| FL_lig_12 | -2.17  | -1.699    | 21.72      |
| FL_lig_15 | -1.03  | -1.258    | 22.154     |
| FL_lig_16 | -1.45  | -1.687    | 16.316     |
| FL_lig_22 | -0.42  | -0.632    | 50.509     |

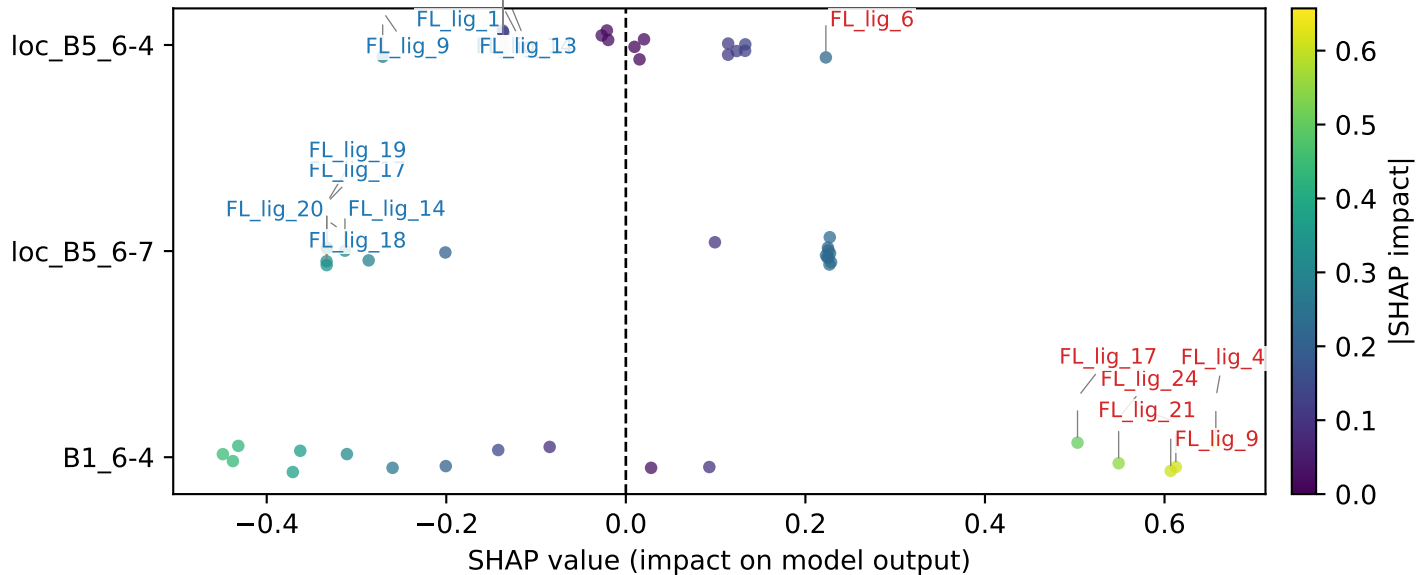
Distribution of Selected Features



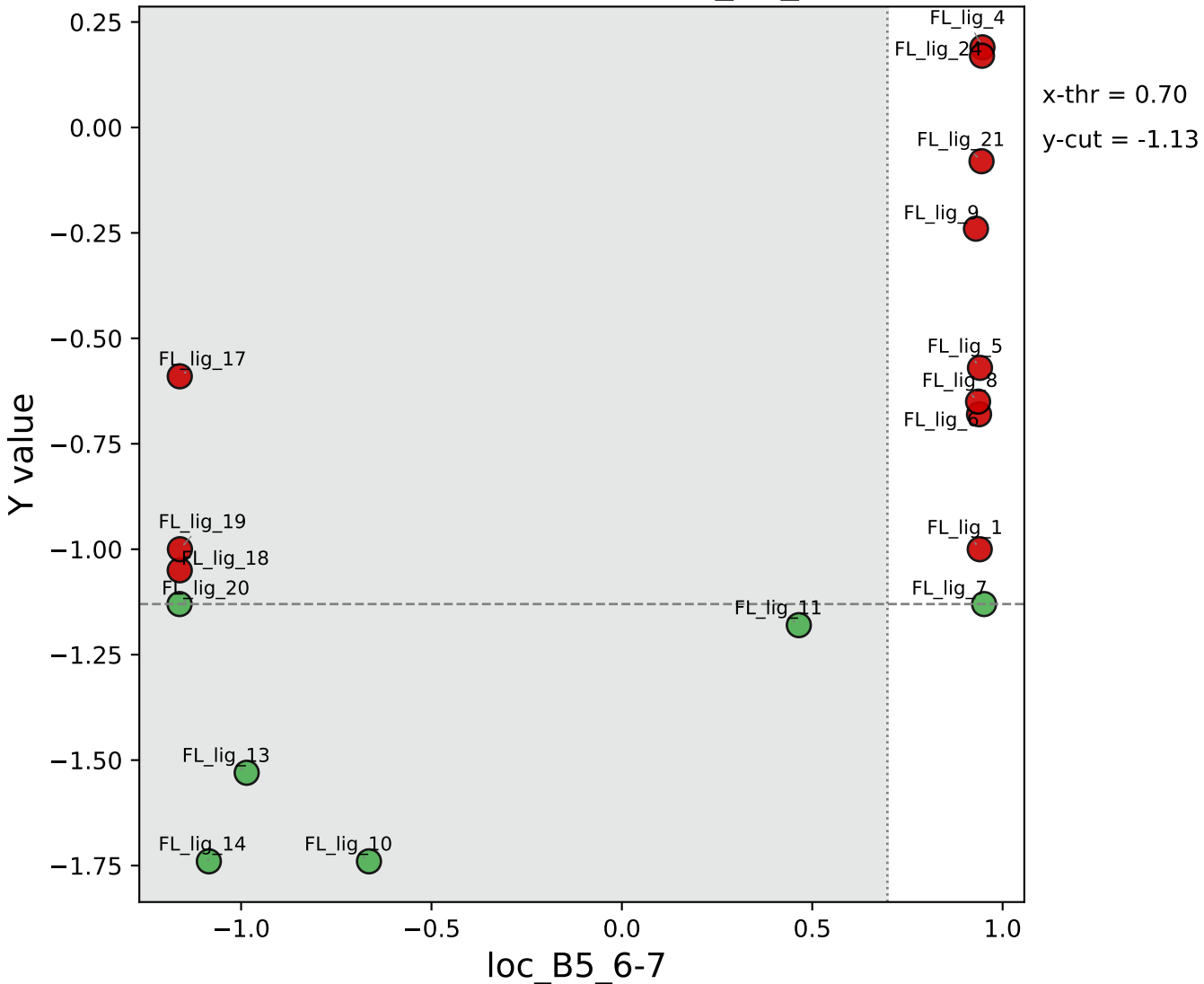
Descriptive Statistics

| Feature    | mean   | std   | min    | 25%    | 50%   | 75%   | max   |
|------------|--------|-------|--------|--------|-------|-------|-------|
| loc_B5_6-7 | 0.092  | 1.001 | -1.162 | -1.085 | 0.93  | 0.94  | 0.951 |
| B1_6-4     | 0.198  | 1.087 | -0.964 | -0.741 | -0.17 | 1.501 | 1.9   |
| loc_B5_6-4 | -0.034 | 0.969 | -2.03  | -1.039 | 0.037 | 0.806 | 1.608 |

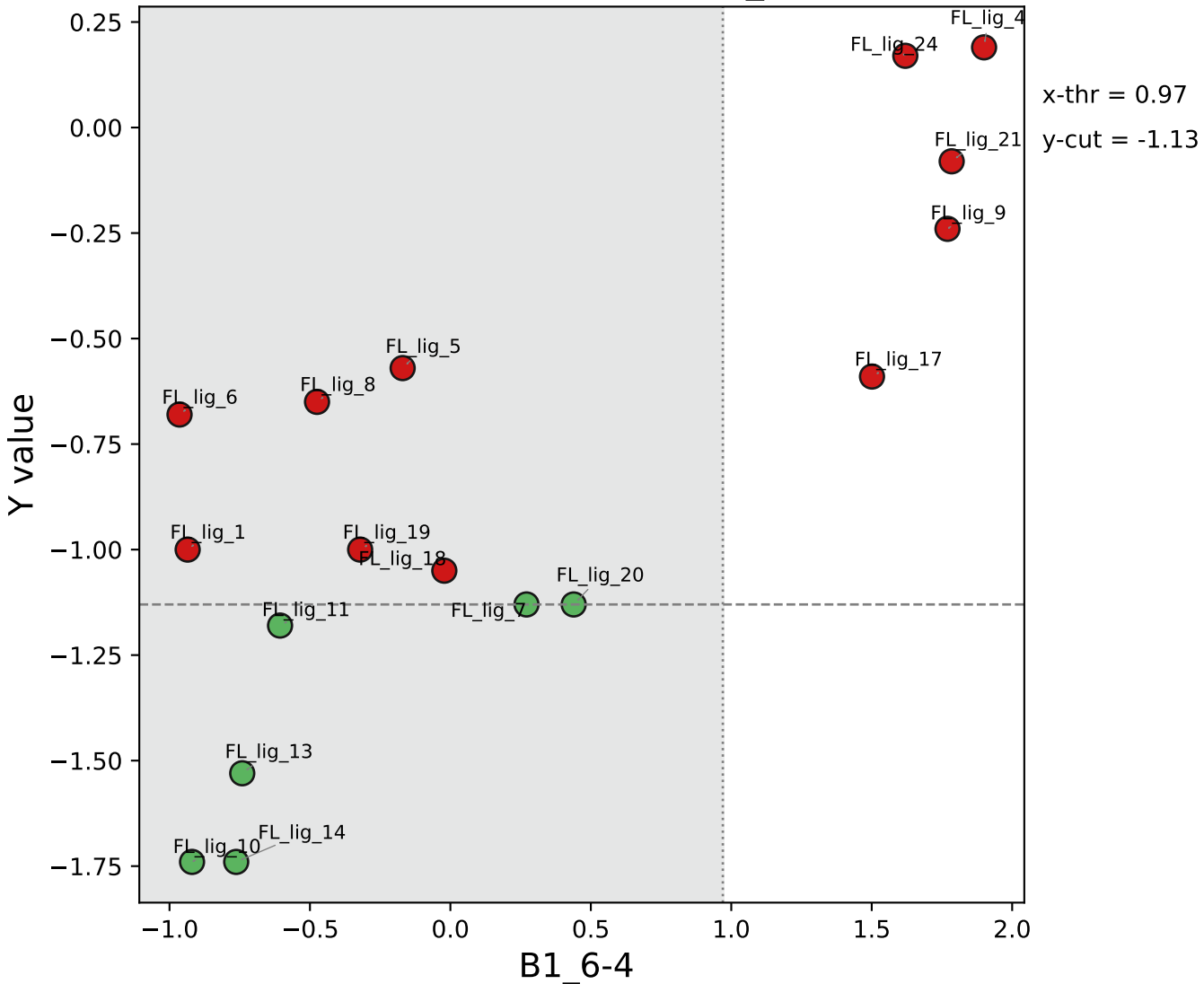
SHAP beeswarm — top 3 features (target)



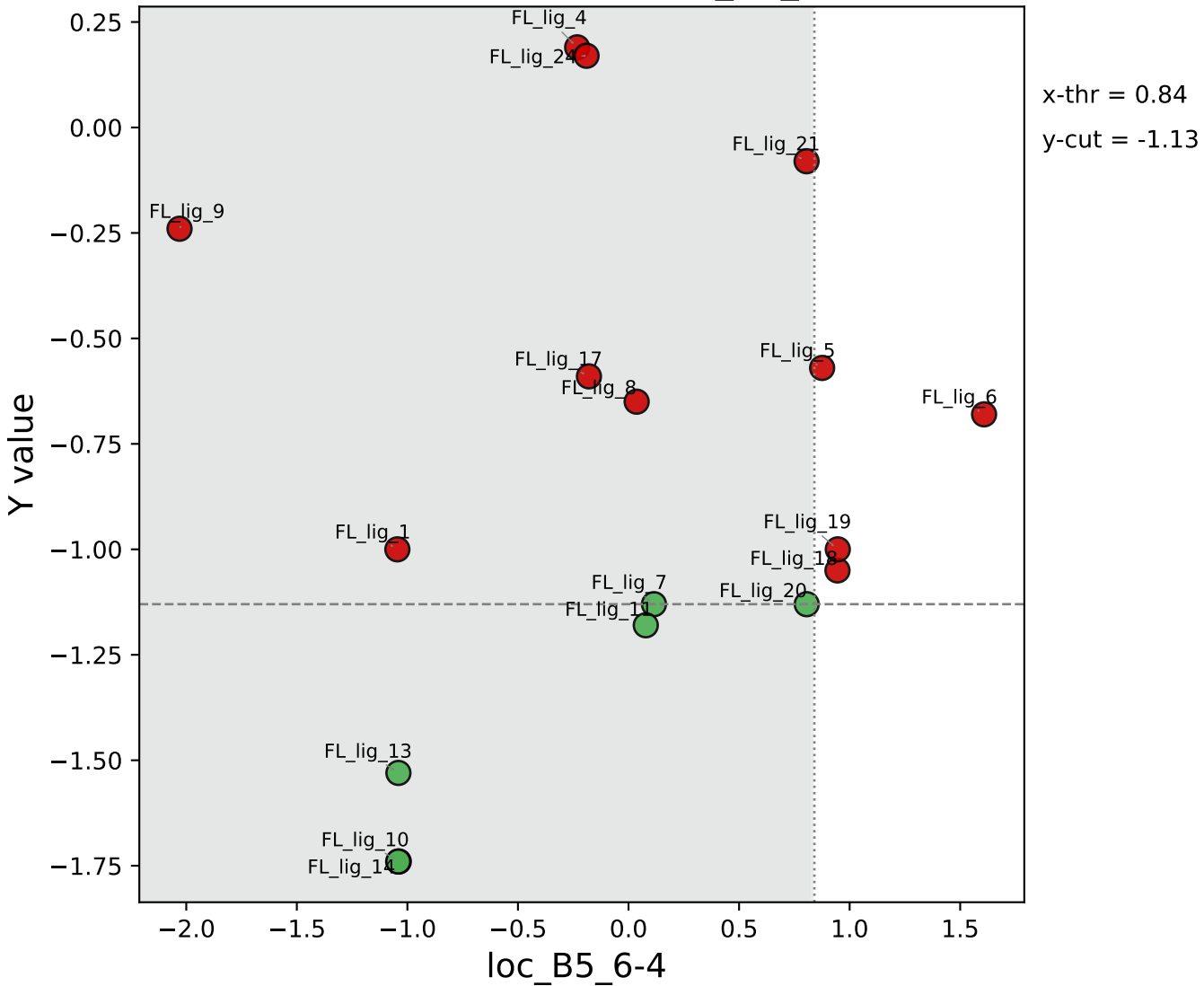
# Threshold analysis: loc\_B5\_6-7



# Threshold analysis: B1\_6-4

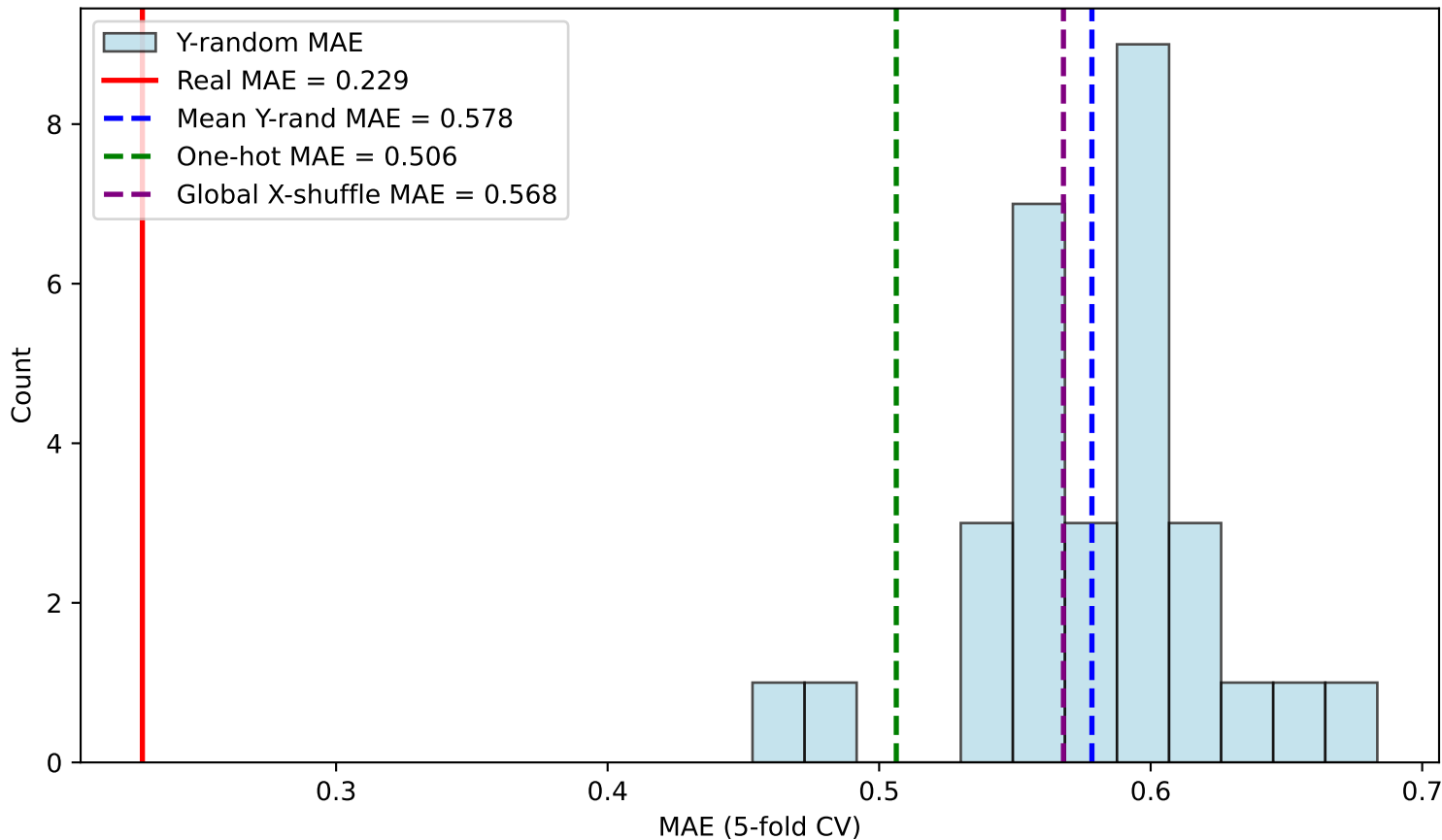


# Threshold analysis: loc\_B5\_6-4

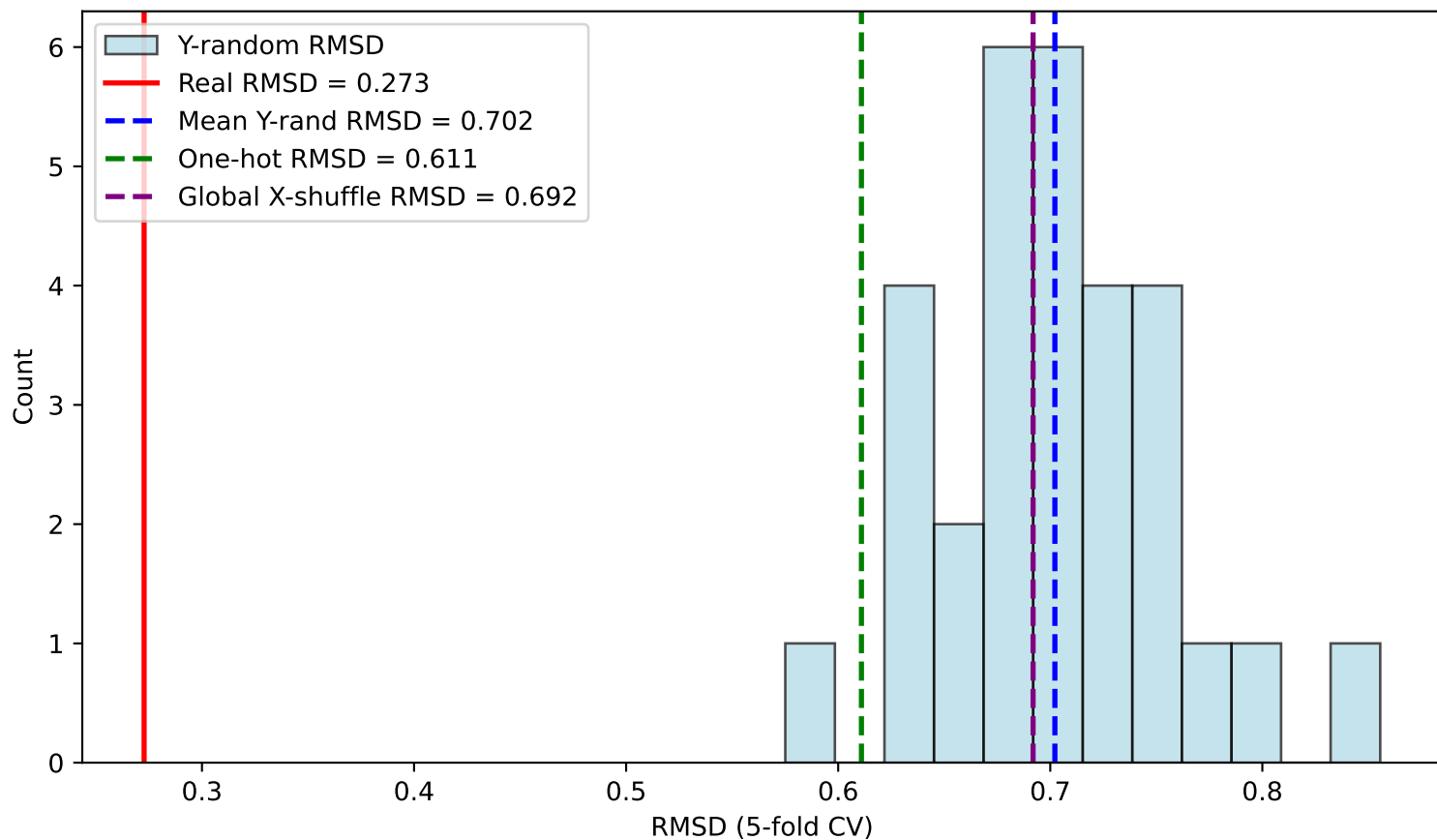




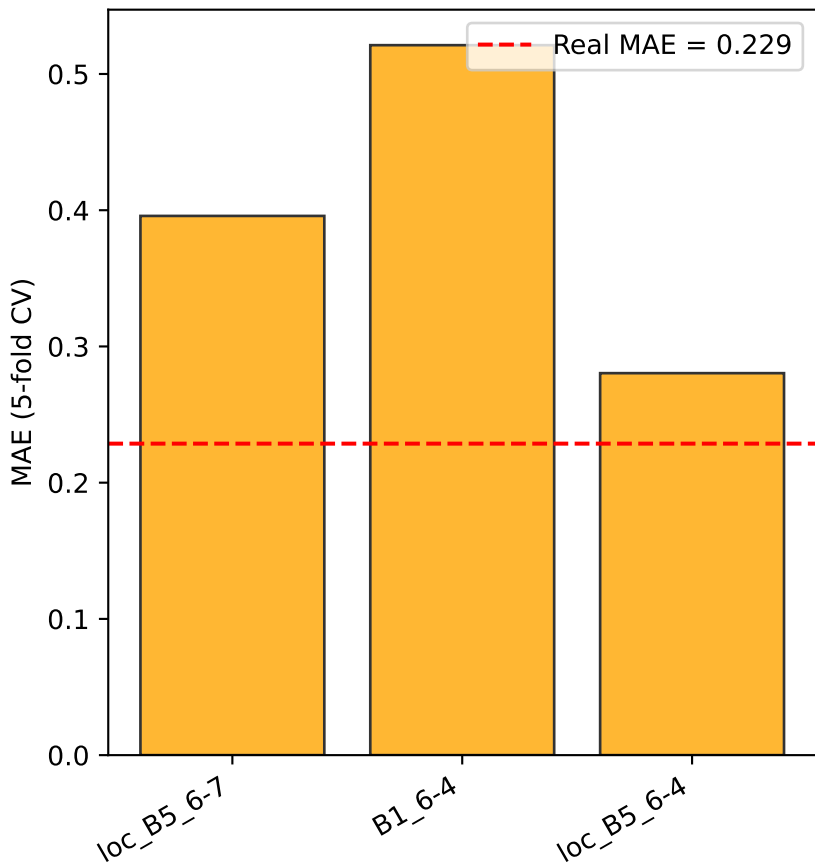
Validation (lower is better): MAE



Validation (lower is better): RMSD



Per-feature shuffle test: MAE (lower is better)



Per-feature shuffle test: RMSD (lower is better)

